

# Triplet transport on M22 cap geometry

Exact transport obstructions, an orientation-cover alignment  
interaction and certified quark residual gaps

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Mathematical results with reproducible certificates. Publication novelty and the particle-physics interpretation require independent review. The nominated golden CKM coefficient and 66-degree phase are not derived.

Research manuscript and continuation of the protected-flavor monograph. Prepared 5 October 2026. Status: computationally certified mathematical results with proofs; publication novelty and the proposed particle-physics interpretation require independent review. No golden CKM coefficient, CKM phase, or experimental mixing anchor enters the group, connection, or overlap enumeration.

## Abstract and the claim being tested

We reconstruct a canonical decorated connection on a 7,392-cap M22/A5 geometry recovered from the Wilson work. Its order-five scalar adjacency  $B$  satisfies  $B^2 = X + 6I$  in the preserved 152-dimensional commutant, so it cannot supply eigenframes independent of  $X$ . The cap graph consists of 176 distance-regular components with 42 vertices, with intersection array  $\{6,5,1;1,1,6\}$ . Each cap has five A4 subgroups. Matching them across a common D5 by their common involutions defines a parameter-free five-letter edge permutation. The resulting holonomy is S5. On the exterior square of the standard four-dimensional hyperplane representation, every edge exchanges the two local golden A5 triplets. There is no parallel rank-three projector on the base graph. Its connected 84-vertex orientation cover has A5 holonomy and admits exactly two such projectors; a positive interaction selects them. Independently, an exact census of the genuine complex triplet of 3.A6 gives 151 complete Abelian residual frames and 111 squared entry values. Every nonzero entry has magnitude at least 0.0759431758699. Two involution residuals also have an exact fixed-entry obstruction. These results identify concrete failures and a precise alignment repair, rather than deriving the nominated golden CKM relation.

The central question is whether the recovered symmetry can enforce  $C_{\text{eff}} = |V_{ub}|(1-|V_{ub}|^2)/(|V_{us}||V_{cb}|) = \phi^{(-2)}$  and a physical CKM phase of 66 degrees. The results below do not establish either equality. In particular, geometric transport, a representation's character field, a central scalar phase, and the physical CKM invariant are different objects.

## Exact scalar identity and the 42-cap component

$$B^2 = X + 6I, \quad B^3 = 4X + 11B + 6D$$

The trusted source is `galois_merge/vendor/forge/DATA/RELATION/TASK_SELECTED_WORD_PR OGRAMS_V23/exact_prefix_1023.npz`, SHA256

1351e514fcb5f23832f65665e0a12d2e172f8131c34875de0e03cddc5aec3093. It records the original integral Hecke coordinates of 1, X, Y and their binary words. The original X is orbital A49, and the original Y is -A12-A48-A59+A95+A116+A125. The sole valency-six orbital is B = A134; the sole valency-five orbital is D = A139. Solving against 152 independent preserved words and checking all 511 available extensions reconstructs multiplication by B and D without CKM input.

**Theorem 1.** In the full reconstructed regular commutant representation,  $B^2 = X + 6I$  and  $B^3 = 4X + 11B + 6D$ . Consequently  $B^4 - 4B^3 - 17B^2 + 24B + 36I = 0$ . Its regular characteristic polynomial is  $(t-6)^9(t-2)^{65}(t+1)^{27}(t+3)^{51}$ . These are exact integral matrix identities, not rounded spectral matches.

**Proof.** Express the orbitals B and D in the selected exact word basis using rational matrix inversion. Their recovered multiplication matrices have integral entries. Direct integer matrix products give the stated identities. The characteristic polynomial is computed independently from the integral matrix. The complete matrices and replay script are supplied with the receipts. The four distinct roots occur, so the displayed quartic is the minimal polynomial.

An independent degree-22 permutation construction uses the ATLAS standard M22 generators. A subgroup generated by an involution and an element of order three with product of order five has order 60. The selected cap class has subdegree counts  $1^1, 5^1, 6^1, 10^1, 15^2, 20^4, 30^{42}, 60^{100}$ , matching the recovered action. Enumerating its cosets accounts for all 443,520 group elements and 7,392 caps. Each cap contains six C5 subgroups; a C5 lies in exactly two caps in this class. The corresponding undirected cap graph has valency six. The implementation finds a 42-vertex component with distance layers 1,6,30,5 and the stated intersection array; transitivity gives 176 components.

The component graph is a known Hoffman-Singleton second-subconstituent graph. Its spectrum is  $6^1, 2^{21}, (-1)^6, (-3)^{14}$ . Neither the graph nor its intersection array is claimed as a new discovery. The five distance-three neighbors together with a vertex form an antipodal six-cap block; the component has seven such blocks. The exact component stabilizer acts faithfully as all 2,520 even permutations of the seven blocks, identifying it as A7. A six-cap block stabilizer acts faithfully as all 360 even permutations of its six caps, identifying an ordinary A6 containing the cap A5 with index six.

The identity  $B^2 = X + 6I$  means every X eigenspace is B-invariant and conversely B selects only branches already lying in these spaces. Since the recovered X eigenvalues correspond to four distinct  $B^2$  values, B and X have the same eigenspace decomposition. The order-five scalar operator therefore does not repair the earlier failure to obtain a new three-family mixing frame.

## A canonical decorated transport

$$\Omega = \frac{EG_2}{\sqrt{5}}, \quad \Omega^2 = I, \quad \Omega T = -T\Omega$$

Each A5 cap has five A4 subgroups, equivalently the normalizers of its five Klein four subgroups. Adjacent caps intersect in D5. Each local A4 meets this D5 in a single order-two subgroup, and the five A4 subgroups exhaust its five involutions. Match the source and target A4 subgroups with the same actual involution. This defines a unique bijection of the five local letters, hence a permutation edge transport. The construction is covariant under M22 conjugation and has no continuous coupling or chosen CKM angle.

**Theorem 2.** For this incidence-matching connection, every edge permutation is odd and the holonomy of a 42-cap component is S5. On  $W = \text{exterior-square } R$ , where R is the four-dimensional standard hyperplane representation of the local five letters, transport exchanges the self-dual and anti-self-dual three-dimensional subspaces.

**Proof.** The supplied cap permutations identify all local subgroups and all 252 directed edge transports by exact set intersections. Every edge is odd. Products around loops, after a spanning-tree trivialization, generate all 120 permutations of five letters. An independently written verifier recomputes every transport from conjugated A4 and cap intersections, uses a depth-first rather than breadth-first tree, and again obtains the entire S5. One explicit five-cycle is cap 0, cap 1083, cap 2333, cap 3074, cap 5702, cap 0. Its loop permutation is (2,0,4,3,1) in zero-based image notation, an odd permutation. Vertex labels refer to the supplied deterministic coset enumeration.

For the representation-theoretic statement, use the hyperplane basis  $e_{i-e_5}$ ,  $i=1,\dots,4$ , with metric  $G = I + 11^T$ , of determinant five. Let  $G_2$  be its induced exterior-square metric and E the matrix of wedge pairing in the six ordered bivector coordinates. The Hodge operator is  $\Omega = EG_2 / \sqrt{5}$ , and  $\Omega^2 = I$ . Its two projectors  $(I \pm \Omega)/2$  have rank three. For any five-letter permutation p, its induced bivector transport T satisfies  $\Omega T = \text{sign}(p) T \Omega$ .

Every edge is odd, hence maps one Hodge branch to the other. The local A5 preserves the branches. Their order-five characters are  $(1+\sqrt{5})/2$  and  $(1-\sqrt{5})/2$  in opposite classes, identifying the familiar two real golden triplets. This is a real representation-theoretic branch exchange, not a CP phase calculation.

For an independent irreducibility check, the bivector character is  $\chi(p) = ((\text{fix}(p)-1)^2 - (\text{fix}(p^2)-1))/2$ . Its squared character sum is 120 over S5, giving inner product one, and is 120 over A5, giving inner product two. The full S5 commutant is therefore scalar, while the A5 commutant is spanned by I and Omega. Exact commutator-rank calculations reproduce dimensions one and two.

Corollary. There is no parallel Hermitian rank-three projector on this connected base connection. Any such projector would commute with every loop at the root. The irreducible S5 action forces it to be a scalar idempotent, hence either zero or identity, of rank zero or six. The theorem excludes parallel projectors for the specified connection; it does not prohibit homogeneous A5 triplet bundles or all possible M22 models.

## What edge symmetry actually enforces

$$\dim \text{Hom}_{D_5}(3, 3^t) = 1, \quad \dim \text{Hom}_{D_5}(3, 3^{\bar{t}}) = 2$$

The D5 intersection reveals an additional coefficient issue. The restriction of one golden triplet to D5 is a one-dimensional sign axis plus a two-dimensional plane carrying C5 weights  $\pm 1$ ; its Galois partner has the same axis and plane weights  $\pm 2$ . Transport between adjacent caps exchanges the two five-cycle character classes. Taking the exact character inner products over the ten common elements gives dimension one for a same-branch edge intertwiner and dimension two for an opposite-branch edge intertwiner.

Thus a same-branch covariant map has rank at most one. An opposite-branch map can be invertible, but its axis and plane couplings are independent. The particular incidence matching above picks a unitary map on both channels; D5 symmetry alone does not force their relative strengths. It would be incorrect to turn this particular geometric isometry into a general coupling-protection theorem. A particle interaction must explain why it uses the incidence matching rather than the full symmetry-allowed two-channel family.

Multiplying edges by central scalar phases, including phases from a central cover, leaves  $\Omega$   $T = -T \Omega$  unchanged. A phase factor also cancels in  $T P T^{-1}$ , so it cannot repair the parallel-projector obstruction. A noncentral transport or different fiber content could do so; neither is excluded here. No identification of a cover eigenphase with the CKM phase is made.

## An explicit protected alignment interaction

$$J_{v,s} = (-1)^s \Omega_v, \quad P_{\pm} = \frac{I \pm J}{2}$$

The orientation cover has vertices  $(v,s)$ , with  $s=0,1$ , and lifts an edge  $v-w$  to  $(v,s)-(w,1-s)$ . It is connected because the base has odd loops. Each component has 84 vertices. The lifted holonomy consists exactly of the 60 even permutations, A5. The field  $J_{(v,s)} = (-1)^s \Omega_v$  is parallel because edge transport anticommutes with  $\Omega$ . The deck exchange changes the sign of J.

Theorem 3. On this cover the only parallel Hermitian rank-three projectors are  $P_+ = (I+J)/2$  and  $P_- = (I-J)/2$ . For dimensionless Hermitian endomorphism order parameters  $P_a$ , an explicit potential selecting exactly these configurations is the sum of three nonnegative terms: a positive

coefficient  $\lambda$  times the sum of squared norms of  $P_a^2 - P_a$ , a positive coefficient  $\mu$  times the sum of  $(\text{Tr } P_a - 3)^2$ , and a positive coefficient  $\kappa$  times the sum over edges of squared norms of  $P_b - T_{ba} P_a T_{ba}^{-1}$ . The norms use the positive fiber metric.

Proof. Zero energy enforces a rank-three projector and parallel transport. At a root the A5 commutant is  $\text{span}\{I, J\}$ . A Hermitian idempotent in this commutant has eigenvalues  $a+b$  and  $a-b$  on the two inequivalent triplets. Rank three forces these to be 1 and 0 in either order. This gives exactly  $P_+$  and  $P_-$ . Conversely, their parallelism and rank make all terms vanish. On the base the identical constraints have no zero-energy configuration, by the scalar S5 commutant. The cover is therefore a concrete repair of this alignment problem, rather than a choice of a fitted frame.

The selected projectors remain exact for arbitrary positive  $\lambda$ ,  $\mu$  and  $\kappa$  within this stated interaction. This is a classical robustness result for the declared potential family. Symmetry allows additional interactions; no general protection against counterterms or radiative corrections is claimed.

This establishes a local rank-three subspace. It does not by itself yield exactly three light global quark modes, nondegenerate quark masses, a relative up/down frame, a UV origin for the interaction, or a cosmological selection of its two branches. In particular, the two real Hodge sectors are not the positive and negative physical CP vacua. Additional physical fields and a light-mode analysis are necessary before treating the construction as a flavor model.

## Exact complex-triplet residual gap

$$|V_{ij}| \in \{0\} \cup [0.0759431758\dots, 0.9566183482\dots] \cup \{1\}$$

The identified A6 block supplies a concrete place to investigate a genuine complex triplet. Ordinary A6 has no faithful ordinary triplet; its triple cover 3.A6, the Valentiner group  $\text{Sigma}(360 \times 3)$ , does. We use the published generators of Hagedorn, Meroni and Vitale, equations (7) and (33), and close them exactly in  $Q(\sqrt{5}, \omega)$ , with  $\omega^2 + \omega + 1 = 0$ . The closure has 1,080 elements and a center of order three, and its real A5 subgroup has order 60. A consistent embedding of this triple cover into a particular central M22 cover is a separate requirement, not an assumption silently imposed by this calculation.

The complete minimal Abelian residual frames comprise 40 order-three frames, 45 order-four frames, 36 order-five frames, and 30 joint Klein frames, totaling 151. The order-three frame count is smaller than the number of cyclic order-three subgroups because central multiplication gives the same projectors. Exact spectral-projector enumeration checks coverage of all 474 noncentral elements of orders three, four and five. Order-twelve and order-fifteen lifts have the same frames as their cubes; order-two and order-six residuals leave a two-dimensional degeneracy. Higher Abelian residuals add the center or already contain one of the complete frames. This classification agrees with the published subgroup analysis; it is not presented as a newly discovered flavor group.

Theorem 4. Every entry probability between the 151 complete frames belongs to the supplied exact set of 111 values. The smallest positive value is  $p_{\min} = 0.005767365961210174\dots$ , and the largest value strictly below one is  $p_{\max} = 0.9151186641666433\dots$ . Consequently every nonzero magnitude is at least  $0.07594317586992\dots$ , and every subunit magnitude is at most  $0.956618348228092\dots$ . Rephasing and all row and column mass permutations preserve these bounds.

Proof. Construct cyclic projectors as the finite Fourier sums of exact group powers, retaining the three projectors of trace one. Construct Klein projectors by their four characters, retaining the three projectors of trace one. For all 11,476 unordered frame pairs with repetition, evaluate all nine traces  $\text{Tr}(P_i Q_j)$ , giving 103,284 exact comparisons. Computation is in  $Q(\zeta_{60})$ , reduced modulo  $\Phi_{60}$ . All probabilities are exactly fixed by complex conjugation. Convert them to degree-seven rational polynomials in  $t = \zeta_{60} + \zeta_{60}^{-1}$ . Its minimal polynomial is  $t^8 - 7t^6 + 14t^4 - 8t^2 + 1$ , and the physical embedding is its largest real root,  $2 \cos(\pi/30)$ . Rational root isolation and interval Horner evaluation give strictly ordered, disjoint rational enclosures for all 111 values. This proves both extrema and their ordering without using floating point to choose the winner. The 60th root is a common arithmetic field for orders 3,4,5; its appearance is not a prediction of 66 degrees.

Explicitly,  $p_{\min} = -t^{7/10} - t^{6/20} + 11t^{5/20} + 3t^{4/10} - 3t^{3/5} - 9t^{2/20} - t/5 + 1/2$ . The upper endpoint below one is  $p_{\max} = 2t^{6/9} - 4t^{4/3} + 2t^2 + 1/9$ . The full rational polynomial table and certified intervals are in `receipts/m22_interactions/valentiner_exact_gap.json`. A separate unitary numerical census of all 22,801 ordered frame pairs independently reproduces the extrema; 36 mass label permutations per pair correspond to 820,836 labeled pairs, and no labeling changes the entry spectrum.

The nominated values  $|V_{cb}|=0.0411$  and  $|V_{ub}|=0.00352144218276$  fall inside the forbidden nonzero gap. Thus complete Abelian residual symmetries cannot give that CKM hierarchy in this triplet without symmetry breaking. These numbers are used only after the exact spectrum is constructed, as a comparison to the existing nominated model, not as a current experimental global fit. The result excludes the specified exact residual construction; it does not exclude 3.A6 as a broken flavor symmetry.

## Partial residuals and the fixed-line obstruction

$$|V_{ij}|^2 \in \left\{ 0, \frac{3-\sqrt{5}}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3+\sqrt{5}}{8}, 1 \right\}$$

Leaving a two-dimensional degenerate sector in each quark mass matrix permits continuous rotations, so the preceding complete-frame gap does not apply directly. Nevertheless, retaining an involution in both sectors fixes one line in each. There are 45 order-two elements. Their rank-one projectors are  $(I+g)/2$ . Evaluating all 2,025 ordered projector overlaps exactly gives squared fixed entries 0,  $(3-\sqrt{5})/8$ ,  $1/4$ ,  $1/2$ ,  $(3+\sqrt{5})/8$ , and 1. Their magnitudes are 0, 0.309016994..., 0.5, 0.707106781..., 0.809016994..., and 1.

Theorem 5. If both nondegenerate quark Hermitian mass matrices commute with nontrivial involutions in the stated triplet, at least one CKM entry is in this fixed set, irrespective of the two otherwise arbitrary  $U(2)$  rotations and irrespective of any added compatible CP constraint. Distinct fixed lines cannot give the nominated hierarchical CKM matrix. Identical fixed lines give an exactly decoupled family and hence a two-family mixing block. A zero fixed overlap also supplies an exact zero entry. All these cases require symmetry breaking to realize the nominated matrix with nine nonzero entries.

Proof. Commutation with an involution preserves its one-dimensional  $+1$  eigenspace and its two-dimensional  $-1$  eigenspace. Nondegenerate mass eigenstates therefore include the fixed line in each sector. Their overlap is a CKM entry. Rotations confined to the two-dimensional complements do not change this overlap. The exact projector calculation gives the stated list. The nominated matrix's off-diagonal magnitudes are below 0.3090 and its diagonal magnitudes exceed 0.8091; none is zero or one. Adding CP cannot change the representation-fixed line overlap. This argument does not assume a specific CP automorphism or a chosen mixing phase.

## Publication scope and the next physical problem

The result suitable for scrutiny is a conjunction of exact transport obstruction, explicit cover alignment mechanism, and exact quark residual gaps. It is not a derivation of the golden CKM relation. Known ingredients include the graph, the two A5 triplets, the Valentiner generators and residual-subgroup classification, and the Hodge transformation law. The documented contribution is their explicit reconstruction on the recovered cap system, the canonical involution-matching connection with certified holonomy, the scoped interaction repair, and exact gap certificates applied to the nominated quark mechanism. A literature search did not establish novelty of this combination; independent specialist review remains necessary before submitting a novelty claim.

The remaining viable problem is now narrower. A proposed theory must specify a noncentral, charge-compatible interaction that produces three light quark modes, breaks the exact residual obstruction in a controlled way, and correlates the up/down breaking operators after canonical normalization. It must then derive the coefficient  $\phi^{(-2)}$  and a neutral CP observable corresponding to 66 degrees without inserting either target. Central phases alone, scalar order-five adjacency alone, and exact complete or two-involution residual frames cannot supply that result. Small free corrections that merely fit the missing entries would not resolve the original coupling-protection problem.

## Reproduction and sources

Install `python/m22_frame_requirements.txt`. From the repository root run `python python/certify_m22_c5.py`, `python python/develop_m22_transport.py`, `python python/verify_m22_transport.py`, `python python/develop_valentiner_frames.py`, and `python python/certify_valentiner_gap.py`. The first three scripts distinguish the preserved Wilson regular algebra from the independently enumerated cap geometry. The independent verifier reconstructs subgroup intersections directly and does not reuse the connection's cocycle code or centralizer solver. The exact residual-gap script checks its entire field and real embedding using rational intervals. Logs and JSON receipts are included. A cold replay compares JSON outputs byte for byte; the compressed permutation file is validated by its array contents, not by ZIP container metadata. No CKM input enters any of these five computations.

The standard degree-22 generators are from the ATLAS of Finite Group Representations, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M22/>, generator files `gap/M22G1-p22B0.g1` and `gap/M22G1-p22B0.g2`. Thomas Connor and Dimitri Leemans, The subgroup lattice of M22 (2012), <https://leemans.dimitri.web.ulb.be/atlaslat/m22.pdf>, independently documents the relevant A5, A6 and A7 subgroup classes; the selected class is recorded by explicit permutations rather than inferred only from a class name. Claudia Hagedorn, Aurora Meroni and Lorenzo Vitale, Mixing Patterns from the Groups  $\Sigma(n, \phi)$ , arXiv:1307.5308, <https://arxiv.org/abs/1307.5308>, supplies the Valentiner generators and prior residual analysis. The known 42-vertex graph is described in Wei Jin and Cheryl E. Praeger, Normal quotients of diameter at most two of finite three-geodesic-transitive graphs, *Journal of Combinatorial Theory A* 178 (2021), 105349, <https://doi.org/10.1016/j.jcta.2020.105349>. These sources establish antecedents; they do not independently verify the new receipts or confer a publication claim.